

CLIENT SUMMARY REPORT

Customer Shopping Behavior

Understanding what drives customer decisions, loyalty and repeat purchases

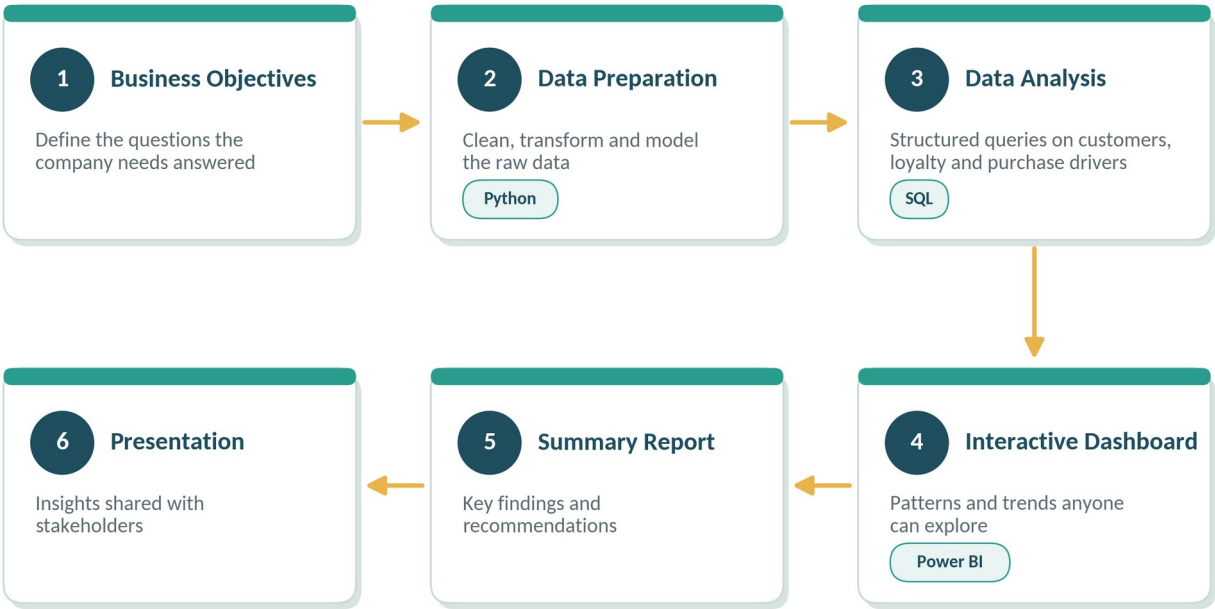
1. Introduction

Thank you for the opportunity to explore your customer shopping data. This report summarizes what we did, what we found, and the steps we recommend to help you grow sales, strengthen customer satisfaction and build long-term loyalty.

We analyzed 3,900 purchases to understand how customers shop across different demographics, product categories and shipping options. The sections that follow walk you through our approach, the analysis, and our recommendations. An interactive Power BI dashboard accompanies this report so your team can keep exploring the results.

2. Our Approach

Our work followed six connected steps, taking your data from raw records to clear, actionable insights:



3. What We Delivered

The project brings together four deliverables, each building on the one before it:

Deliverable	Tool	What It Covers
Data Preparation & Modeling	Python	Cleaning and transforming the raw dataset so it is ready for analysis.
Data Analysis	SQL	Organizing the data in a structured format, simulating business transactions, and running queries on customer segments, loyalty and purchase drivers.
Visualization & Insights	Power BI	An interactive dashboard that highlights key patterns and trends and supports data-driven decisions.
Report & Presentation	Report / Slides	A clear project report with key findings and recommendations, plus a presentation that communicates them visually to stakeholders.

4. Analysis Questions

The analysis is designed to answer these ten questions:

1. What is the total revenue generated by male vs. female customers?
2. Which customers use a discount but still spend more than the average purchase amount?
3. Which are the top 5 products with the highest average review rating?
4. How does the average purchase amount compare between standard and express shipping?
5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.
6. Which 5 products have the highest percentage of purchases with discounts applied?
7. Segment customers into New, Returning and Loyal based on their number of previous purchases, and show the count of each segment.
8. What are the top 3 most purchased products within each category?
9. Are repeat buyers (more than 5 previous purchases) also likely to subscribe?
10. What is the revenue distribution across each age group?

5. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across a range of product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences and subscription behavior, so the business can make smarter strategic decisions.

Dataset Summary

3,900 Rows	18 Columns	37 Missing values (Review Rating)
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The dataset captures three main kinds of information:

Feature Group	What It Includes
Customer Demographics	Age, gender, location and subscription status.
Purchase Details	Item purchased, category, purchase amount, season, size and color.
Shopping Behavior	Discount applied, promo code used, previous purchases, purchase frequency, review rating and shipping type.

6. Exploratory Data Analysis (Python)

We began with data preparation and cleaning in Python, working through the steps below.

Loading and First Look

- **Data loading:** imported the dataset using Pandas.
- **Initial exploration:** used `df.info()` to check the structure and `.describe()` for summary statistics.

Handling Missing Data

We checked for null values and filled the gaps in the Review Rating column using the median rating of each product category.

GOOD TO KNOW

Missing ratings were replaced with the median for their own category, not the overall median. This keeps each product group's ratings more realistic.

Cleaning and Feature Engineering

- **Column standardization:** renamed columns to snake case for readability and easier documentation.
- **Age groups:** created an `age_group` column by binning customer ages.
- **Purchase frequency:** created a `purchase_frequency_days` column from the purchase frequency data.

Data Consistency Checks

We confirmed that the `discount_applied` and `promo_code_used` columns contained exactly the same data, so we dropped `promo_code_used` to avoid duplication.

Database Integration

Finally, we connected the Python script to MySQL Workbench and loaded the cleaned data frame into the database, ready for SQL analysis.

7. Data Analysis (SQL)

Using MySQL Workbench, we ran structured queries to answer the key business questions. Each result is shown below its question.

Revenue by Gender

Total revenue generated by male vs. female customers.

	gender	revenue
▶	Male	157890
	Female	75191

High-Spending Discount Users

Customers who used a discount but still spent above the average purchase amount.

	customer_id	purchase_amount
▶	1480	100
	1413	100
	96	100
	1422	100
	1457	100
	1209	100
	194	100
	770	100
	1406	100
	43	100

Average Review Rating

The top 5 products with the highest average review rating.

	item_purchased	Avrage_Product_rating
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.78

Shipping Type Comparison

Average purchase amount for standard vs. express shipping.

	shipping_type	average_purchase_amount	total_purchases
▶	Express	60.48	39067
	Standard	58.46	38233

Spending vs. Subscription Status

Whether subscribed customers spend more than non-subscribers.

	no_of_customers	subscription_status	average_spent	total_revenue
▶	1053	Yes	59.49	62645
	2847	No	59.87	170436

Discounted Purchases

The 5 products with the highest percentage of purchases made with a discount.

	item_purchased	discount_rate
▶	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

Customer Segments

New, Returning and Loyal customers, based on number of previous purchases.

	customer_segment	Number_of_Customers
▶	Loyal	3116
	Returning	701
	New	83

Top 3 Products per Category

The most purchased products within each category.

	item_rank	category	item_purchased	total_orders
▶	1	Accessories	Jewelry	171
	2	Accessories	Sunglasses	161
	3	Accessories	Belt	161
	1	Clothing	Blouse	171
	2	Clothing	Pants	171
	3	Clothing	Shirt	169
	1	Footwear	Sandals	160
	2	Footwear	Shoes	150
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	161

Repeat Buyers & Subscriptions

Whether customers with more than 5 purchases are more likely to subscribe.

	subscription_status	repeat_buyers
▶	Yes	958
	No	2518

Revenue by Age Group

Total revenue distribution across each age group.

	age_group	total_revenue
▶	Young Adults	62143
	Middle-aged	59197
	Adults	55978
	Seniors	55763

8. Business Recommendations

Based on the analysis, these are the actions we recommend:

Recommendation	How to Put It into Action
Boost Subscriptions	Promote exclusive benefits for subscribers to encourage more sign-ups.
Customer Loyalty Programs	Reward repeat buyers to help move them into the Loyal segment.
Review Discount Policy	Balance the sales boost from discounts with healthy margin control.
Product Positioning	Feature top-rated products prominently in marketing campaigns.
Targeted Marketing	Focus efforts on high-revenue age groups and express-shipping customers.
Gender-Focused Strategy	Target male buyers, who bring in more revenue, or offer rewards and discounts to engage more female customers.

Thank you for reading. We hope these insights help turn shopping data into confident, customer-friendly decisions.